Home Lab Report

Enterprise IT Environment Simulation

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Overview

This report documents the setup and configuration of a self-built enterprise IT home lab. The goal was to simulate a real-world IT environment and gain hands-on experience with tools used in actual IT support and administration roles — rather than relying solely on theory.

The lab covers two main areas: on-premise infrastructure using Active Directory and Group Policy, and cloud-based Microsoft 365 administration including Entra ID, Intune, Exchange Online, and SharePoint Online.

Lab Setup

Component	Details
Server	Windows Server 2022 running on a Virtual Machine
Client Machine	Dell Laptop — Windows 10 Enterprise, Domain joined
Host Machine	Personal laptop (Win 11) hosting the VM
Cloud Tenant	Microsoft 365 E3 Trial subscription.

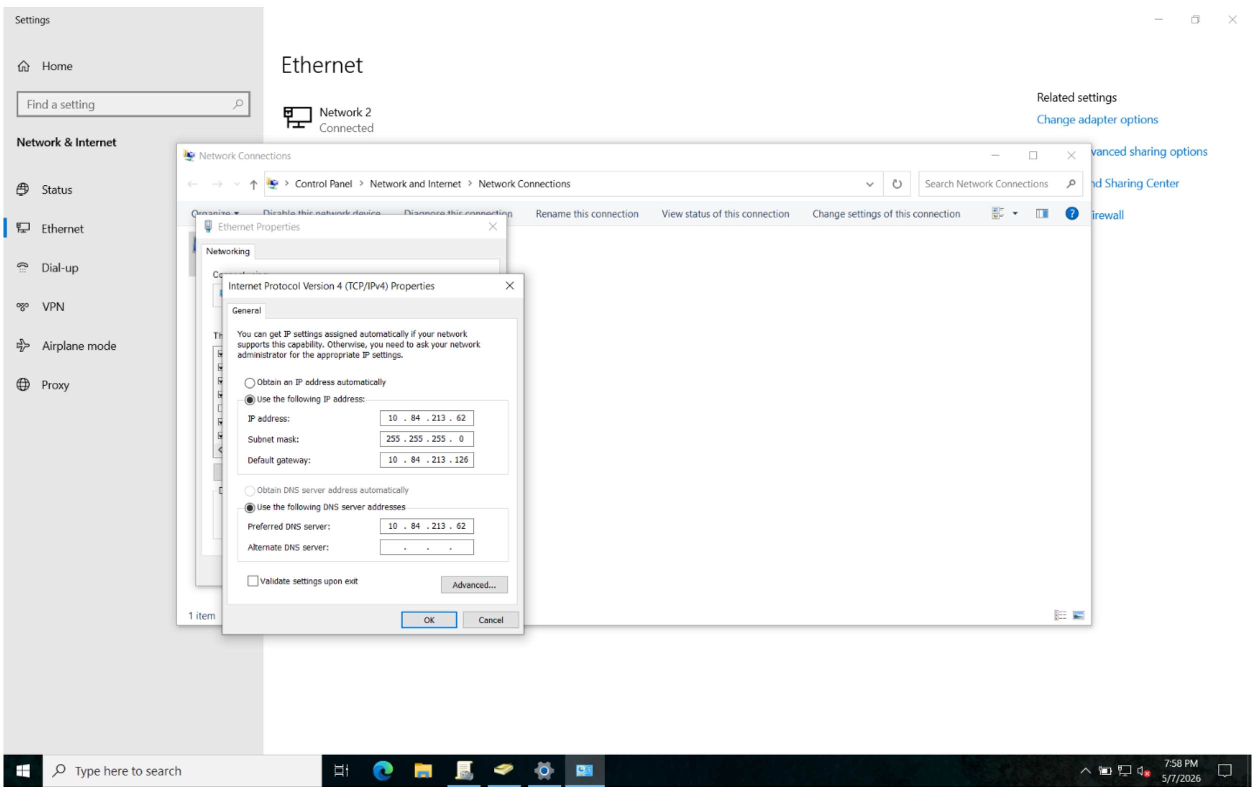
Section 1: Active Directory

This section covers the on-premise Active Directory environment built using Windows Server as a Domain Controller, connected to a Dell laptop running Windows 10 Enterprise as a domain-joined client machine.

1.1 Domain Controller Setup

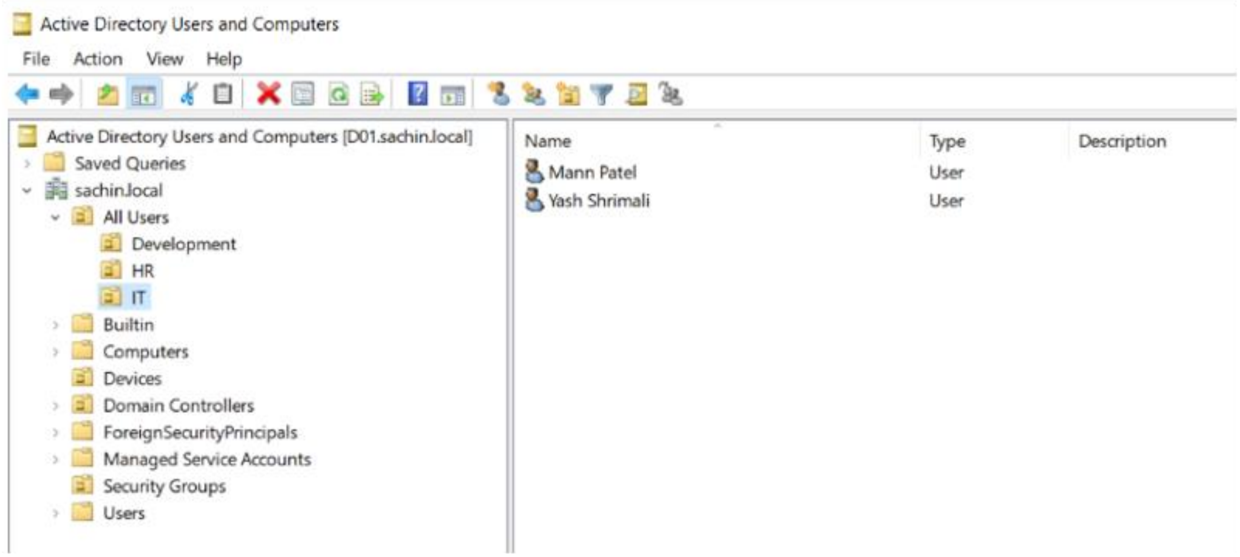
Windows Server was installed on a virtual machine and promoted to a Domain Controller by installing the Active Directory Domain Services (AD DS) role via Server Manager. A new forest and domain were created from scratch.

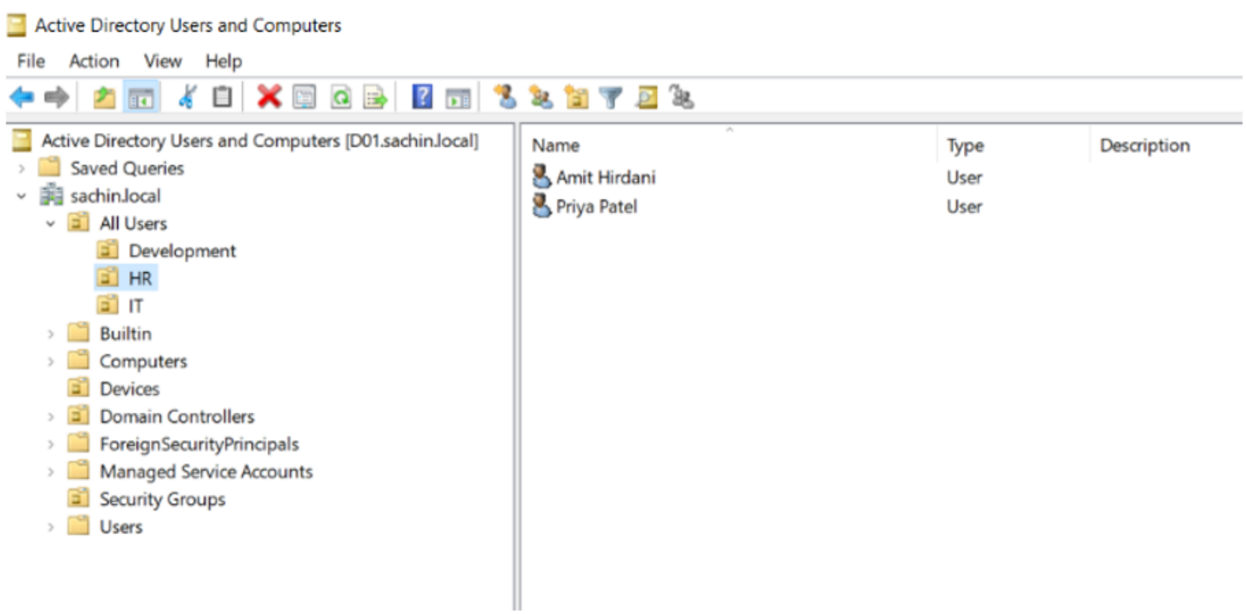
Static IP and DNS configuration on AD Server

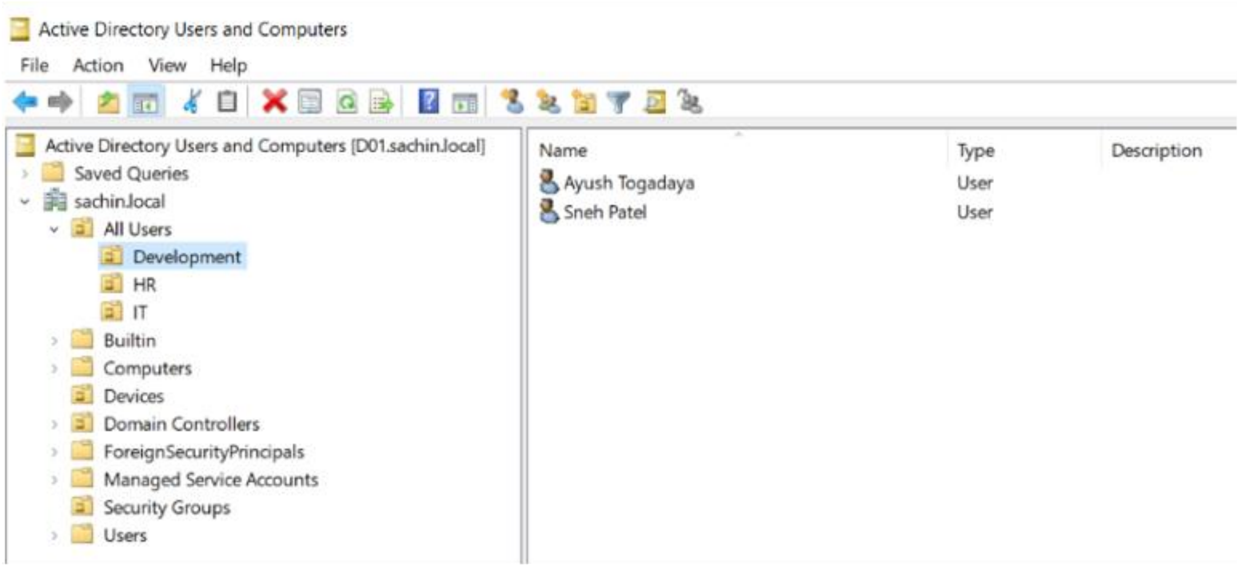


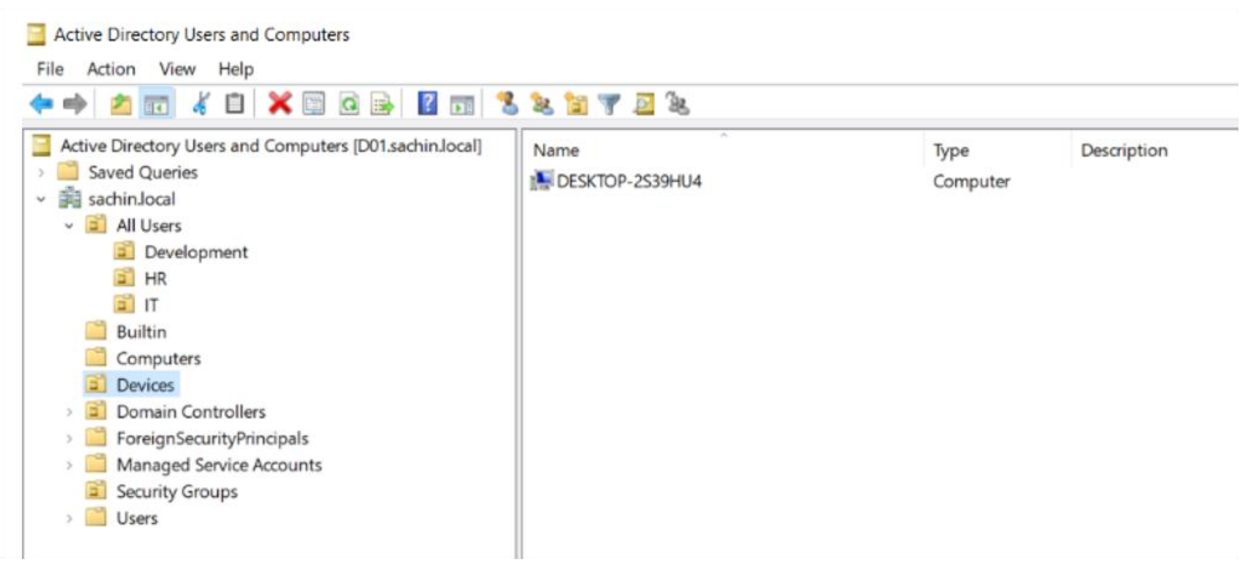
1.2 Users, Computers and Organizational Units

User accounts and computer accounts were created and organized into Organizational Units (OUs) to reflect a real department structure. The Dell laptop was domain-joined and appeared under the Computers OU in Active Directory Users and Computers (ADUC).






Active Directory Users and Computers
File Action View Help
Active Directory Users and Computers [D01.sachin.local]
Saved Queries
sachin.local
All Users
Development
HR
IT
Builtin
Computers
Devices
Domain Controllers
ForeignSecurityPrincipals
Managed Service Accounts
Security Groups
Users
Name
Type
Description
Ayush Togadaya
User
Sneh Patel
User


Active Directory Users and Computers
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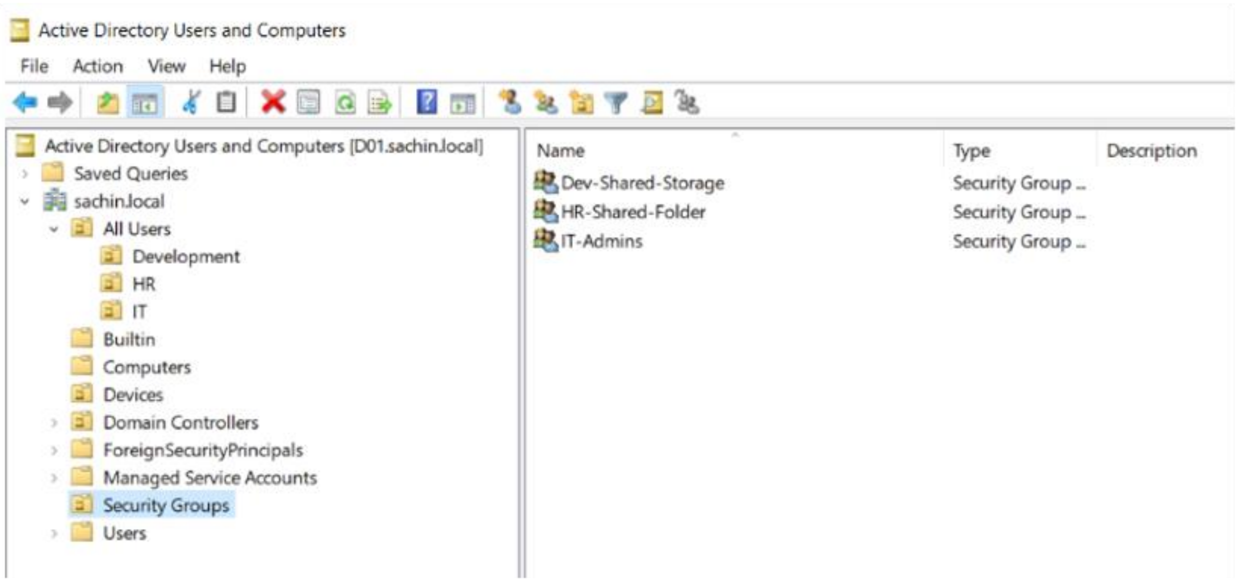
Active Directory Users and Computers [D01.sachin.local]
Saved Queries
sachin.local
All Users
Development
HR
IT
Builtin
Computers
Devices
Domain Controllers
ForeignSecurityPrincipals
Managed Service Accounts
Security Groups
Users
Name
Type
Description
DESKTOP-2S39HU4
Computer

1.3 Security Groups

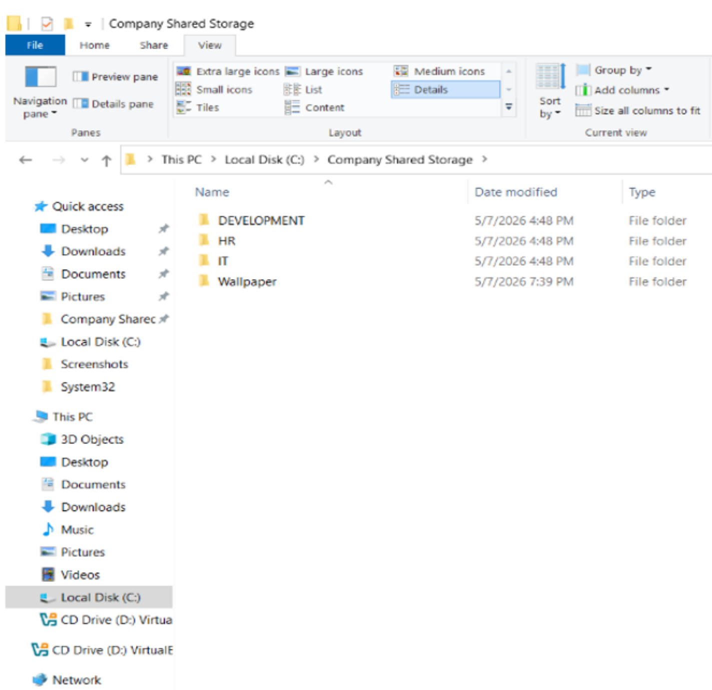
Security groups were created to manage access to resources. Users were added to groups based on their department, and permissions were assigned to the group rather than individual users — reflecting real-world best practice.



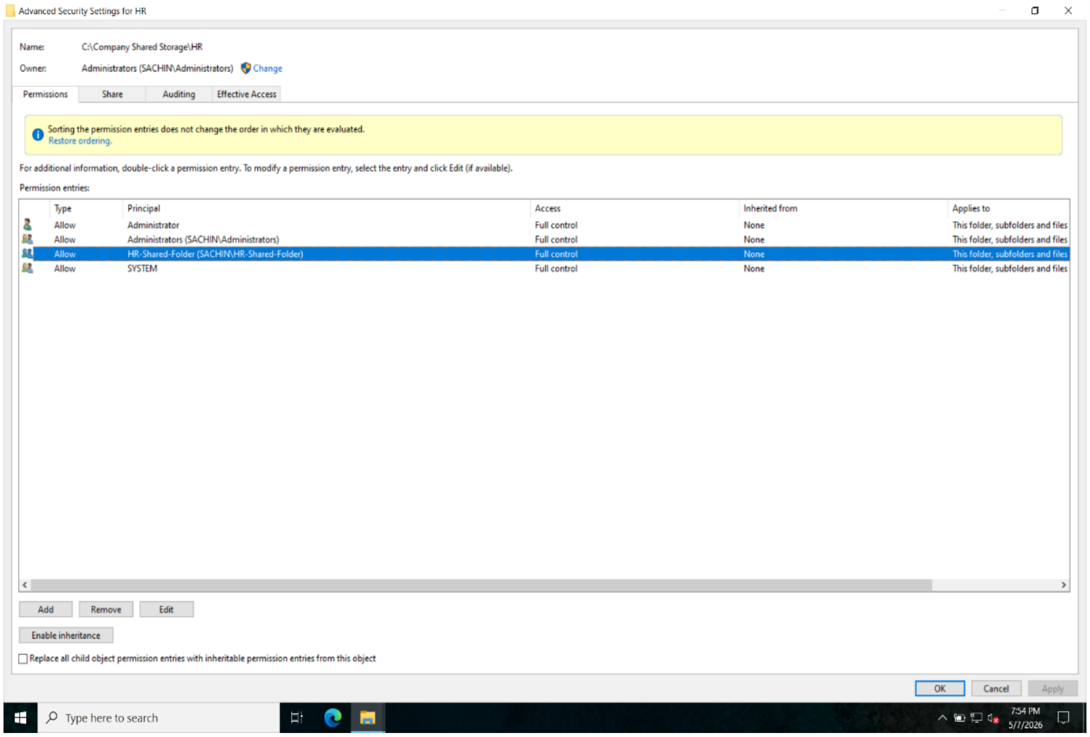
1.4 Shared Folder and NTFS Permissions

A shared folder was created on the server and made accessible to domain-joined machines. NTFS permissions were configured so that different users and groups could access only the folders they were authorized for. Users outside the permitted group could not see or access restricted folders.

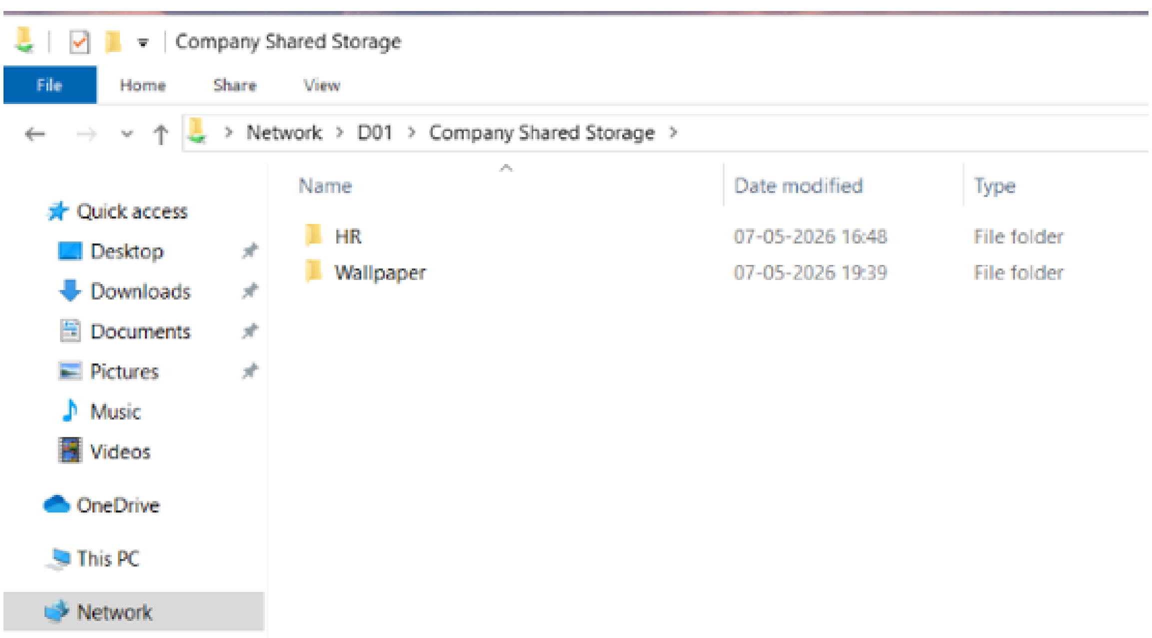
Shared folders



NTFS permission configuration



HR OU users only able to access HR folder and other shared folders.

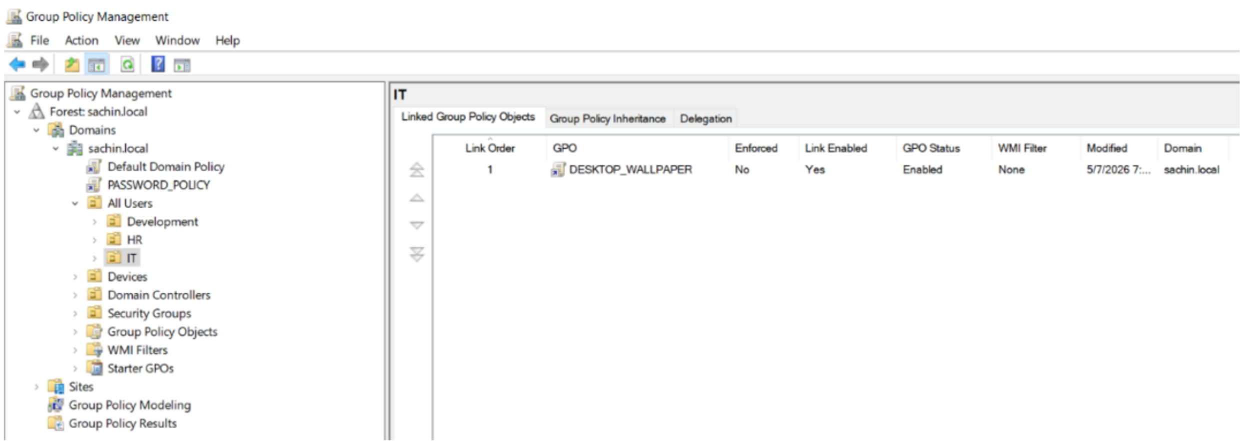


1.5 Group Policy Objects (GPOs)

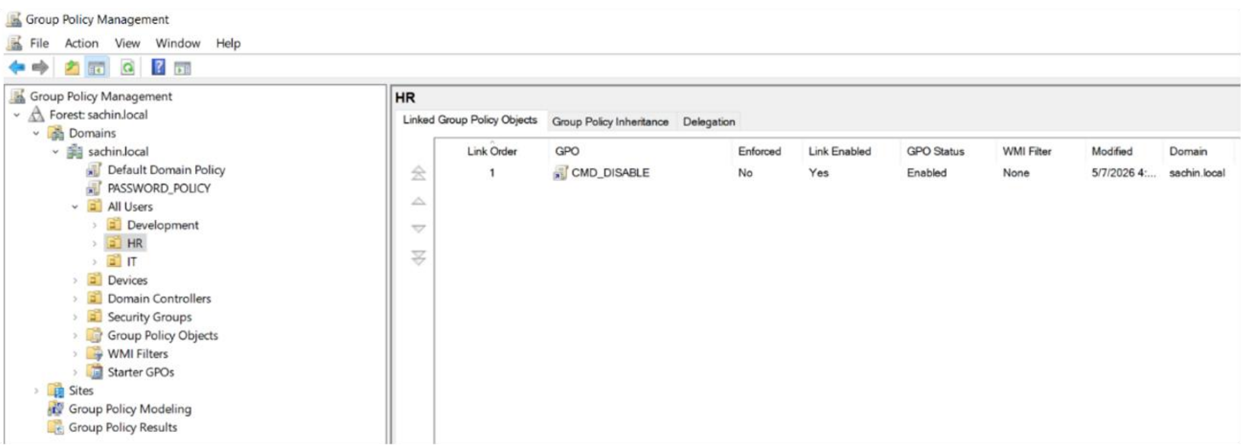
Multiple Group Policy Objects were configured and linked to the domain to enforce settings across all domain-joined machines. The following policies were implemented:

- Wallpaper enforcement — a company wallpaper was pushed to all machines via GPO
- CMD disabled — Command Prompt was restricted for standard users
- Settings app disabled — Control Panel and PC Settings were blocked for standard users
- Software deployment — 7-Zip was deployed automatically to domain machines via GPO software installation

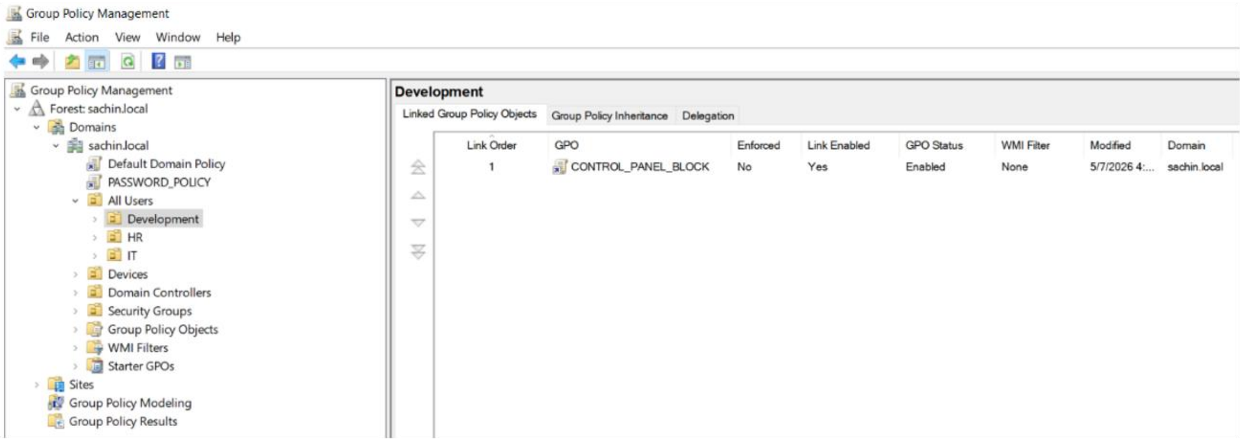
Wallpaper enforcement policy



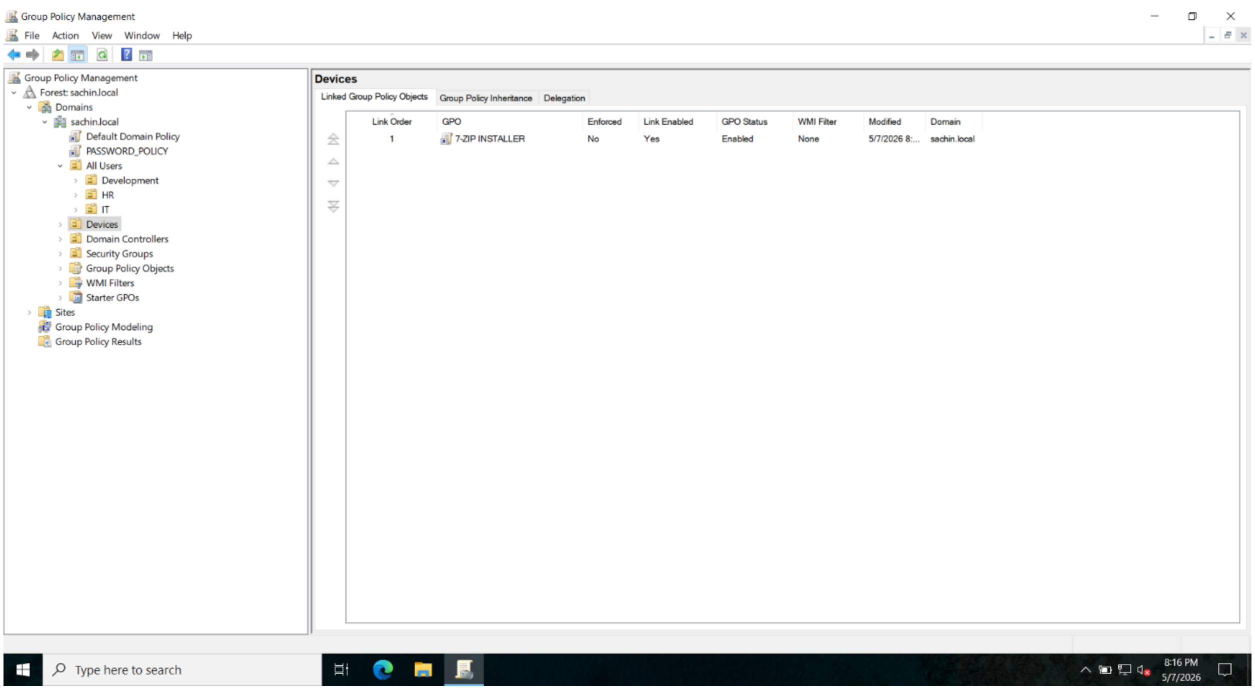
Command prompt disabled



Control Panel Blocked



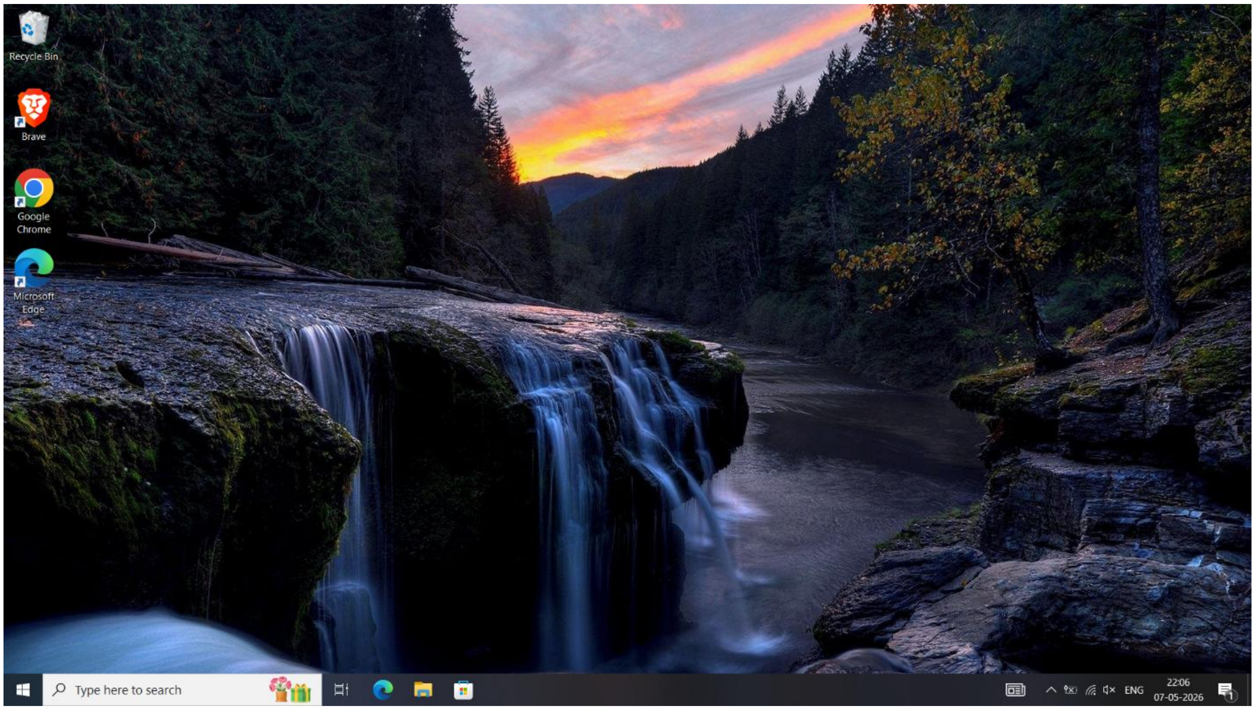
Software installation



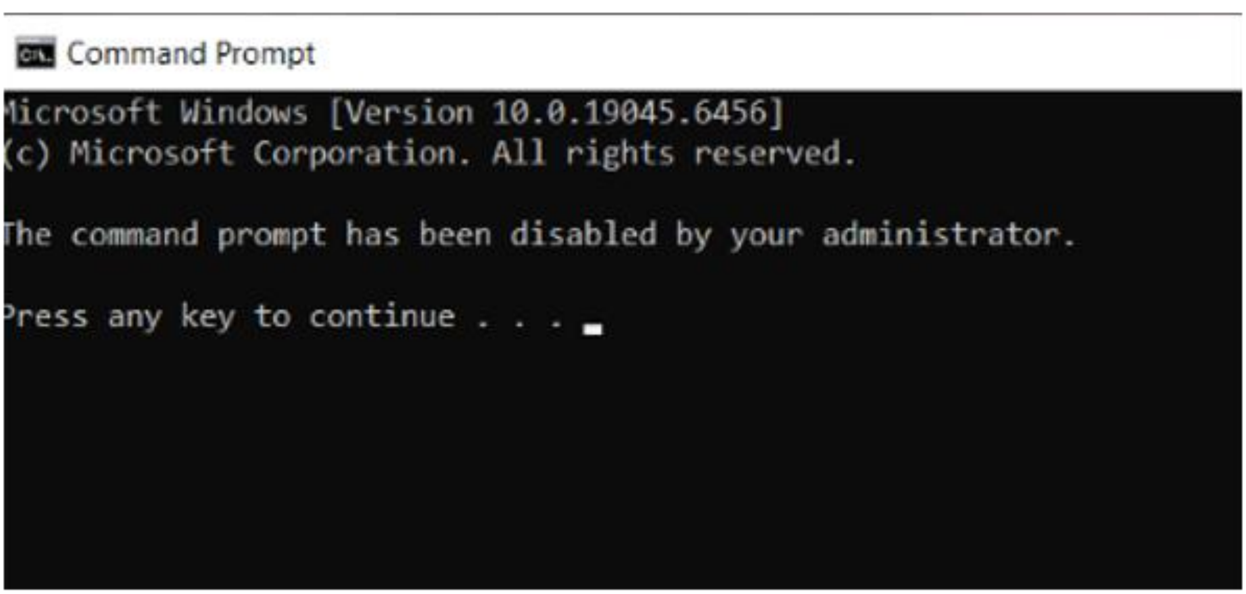
1.6 GPO Results on Client Machine

After GPO was applied, the client Dell laptop reflected all configured policies. CMD and Settings app were inaccessible for restricted users, the enforced wallpaper was applied, and USB storage was blocked. The screenshots below show the result on the client machine.

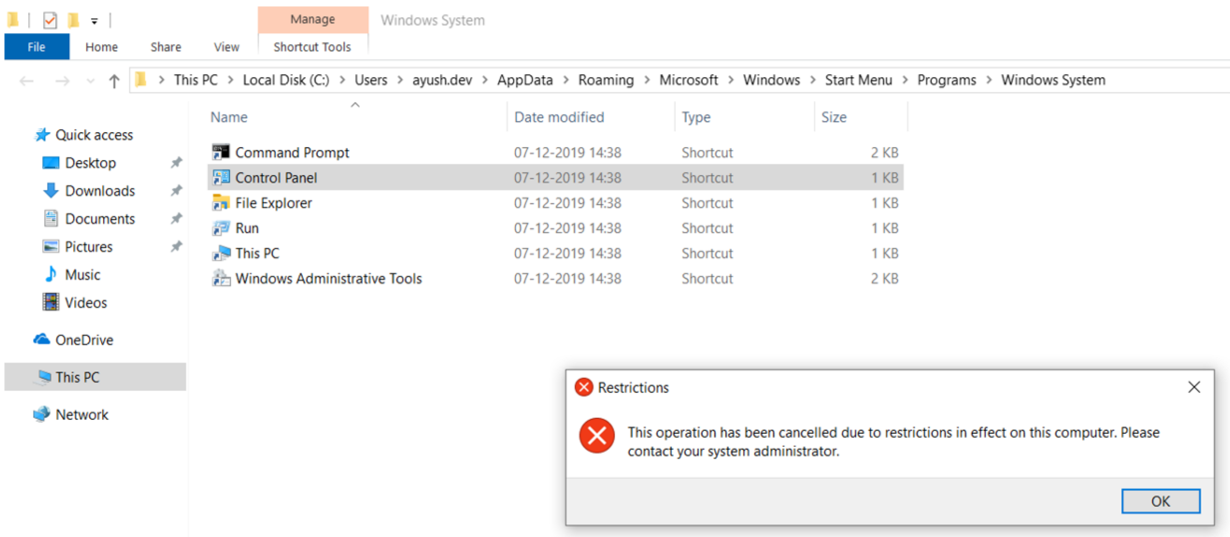
Wallpaper Enforced



Command Prompt Disabled



Control Panel Blocked



Section 2: Microsoft 365 Administration

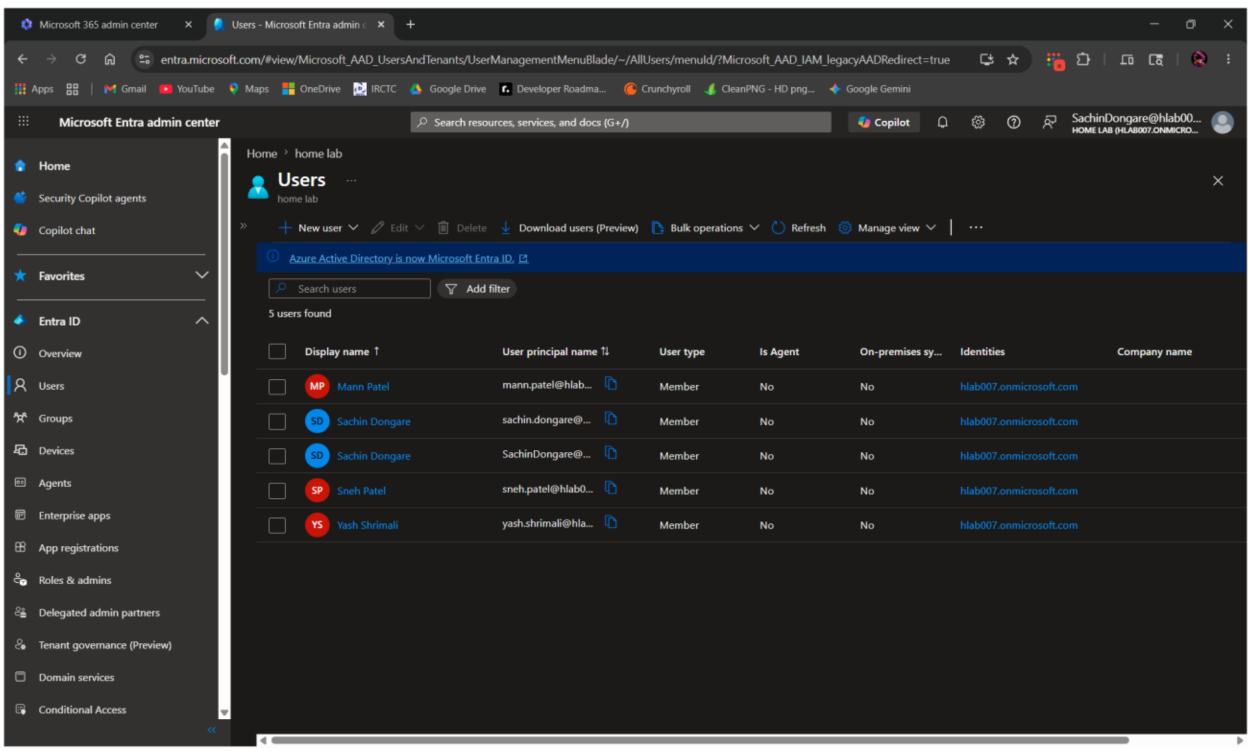
This section covers the cloud-based Microsoft 365 environment configured using a trial tenant. All administration was done through the Microsoft 365 Admin Center, Entra ID, Intune, Exchange Online, and SharePoint Online.

2.1 Entra ID — User and Group Management

Users were created in the Microsoft 365 Admin Center and assigned appropriate licenses. Groups including Security Groups, Microsoft 365 Groups, and Distribution Groups were configured to manage access and collaboration.

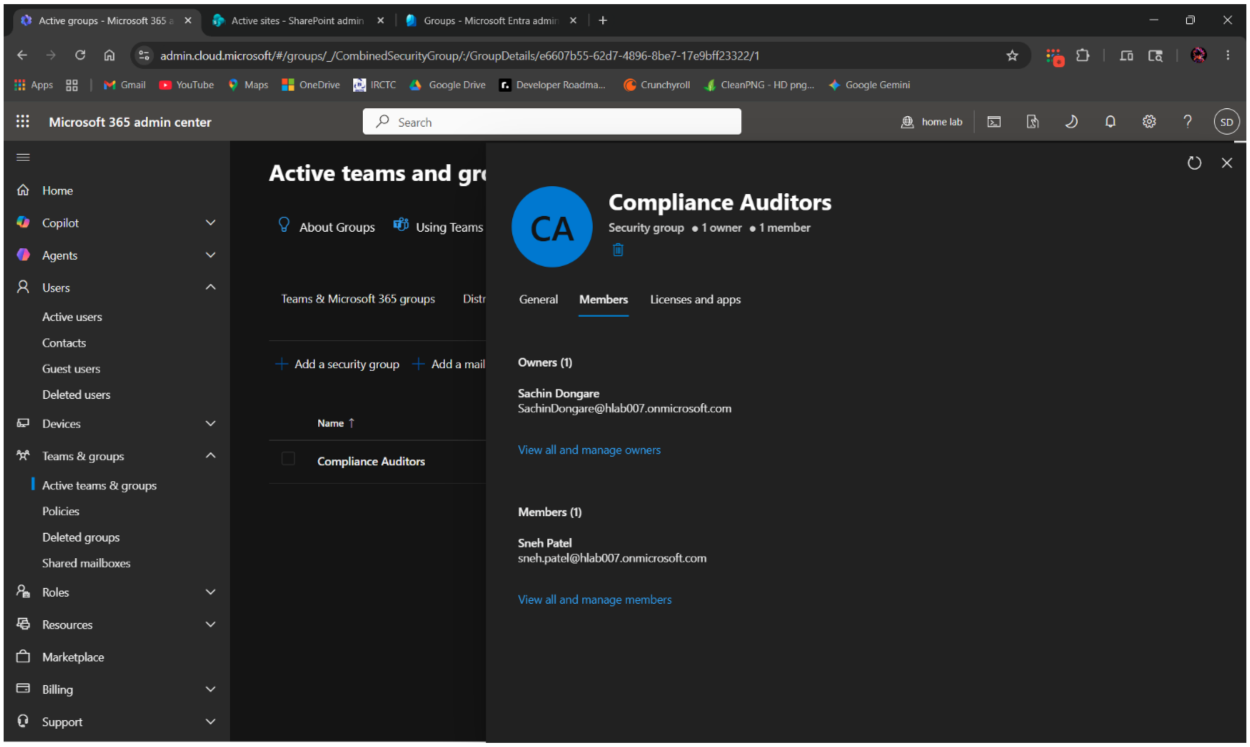
Users and Licenses

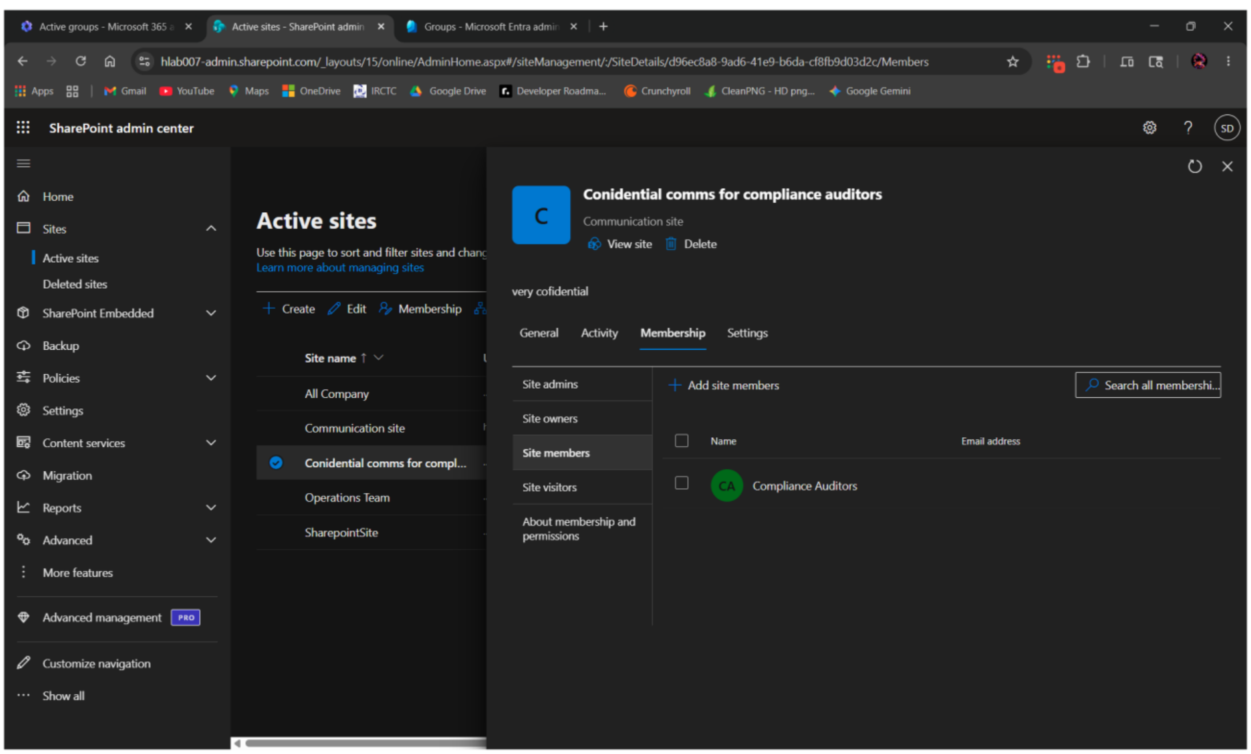
Multiple user accounts were created with licenses assigned. User profiles include department, job title, and contact information — reflecting how users would be set up in a real organization.



Security Groups

Security groups were created and users were added as members. These groups were then used to assign SharePoint site permissions, demonstrating how access control works through group membership rather than individual user assignment.

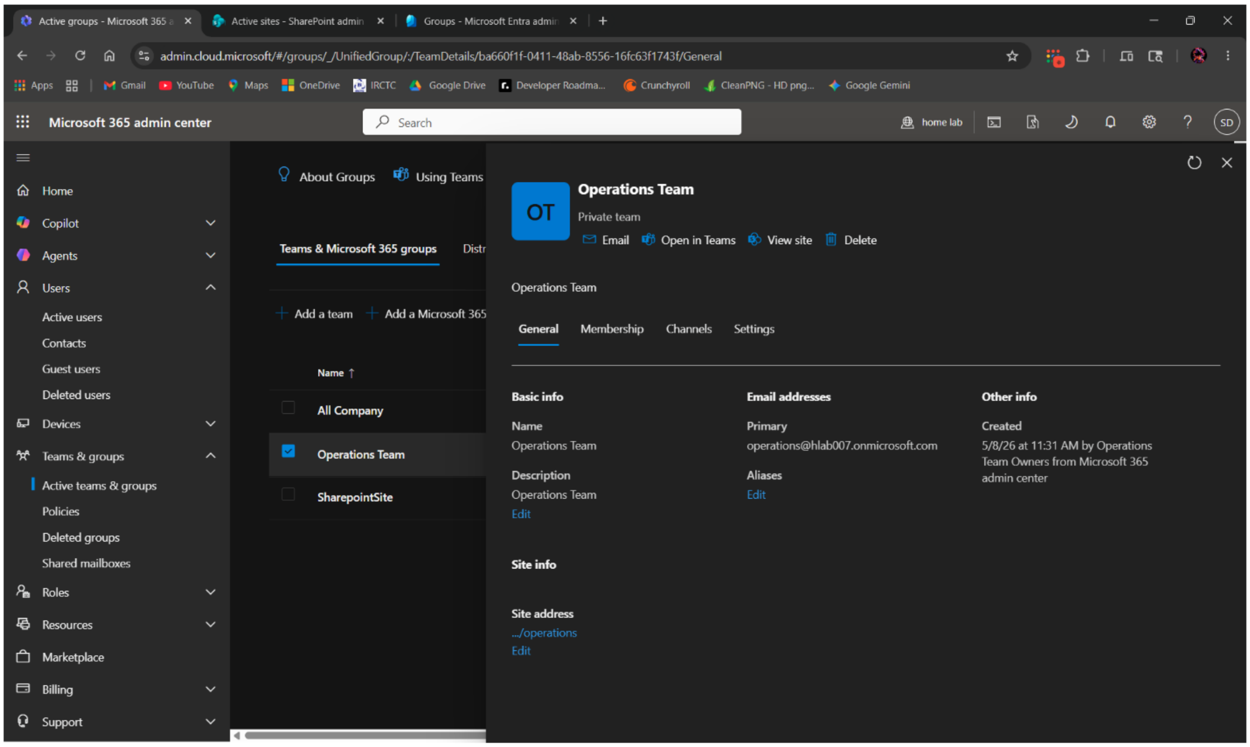




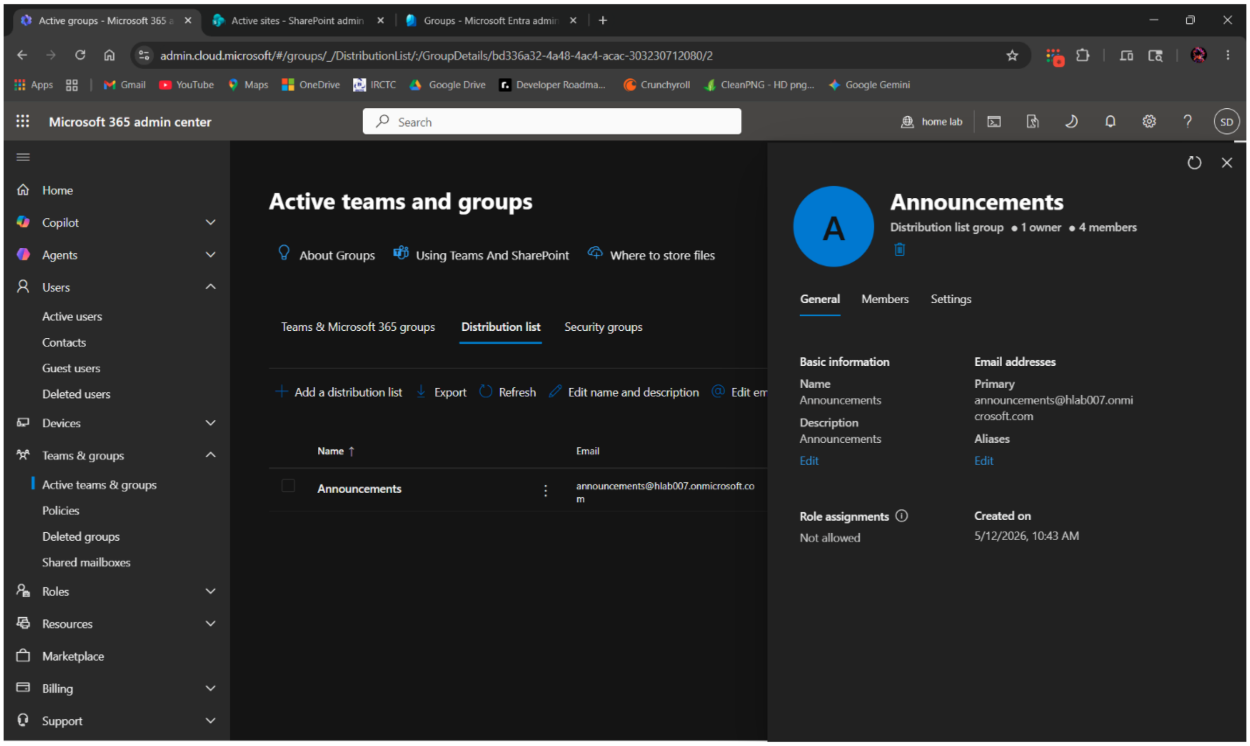
Microsoft 365 Groups and Distribution Groups

Microsoft 365 Groups were created, automatically provisioning a Teams workspace, SharePoint site, and shared mailbox. Distribution Groups were also configured for company-wide email communication.

Microsoft 365 Group

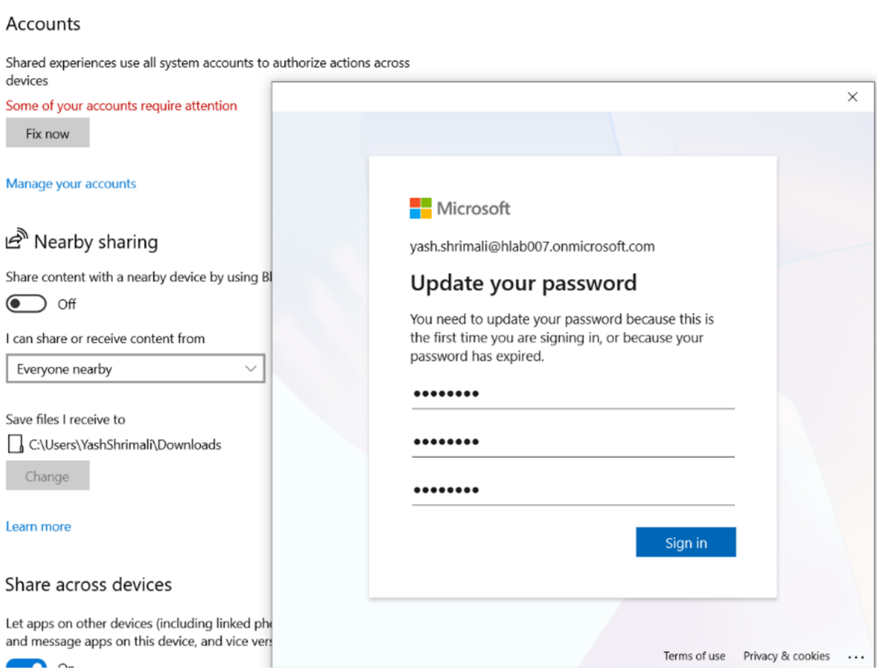


Distribution Group



Session Management

Password reset and session revocation were performed on user accounts. Revoking sessions signs the user out of all devices and apps simultaneously — a key step during offboarding or suspected account compromise.

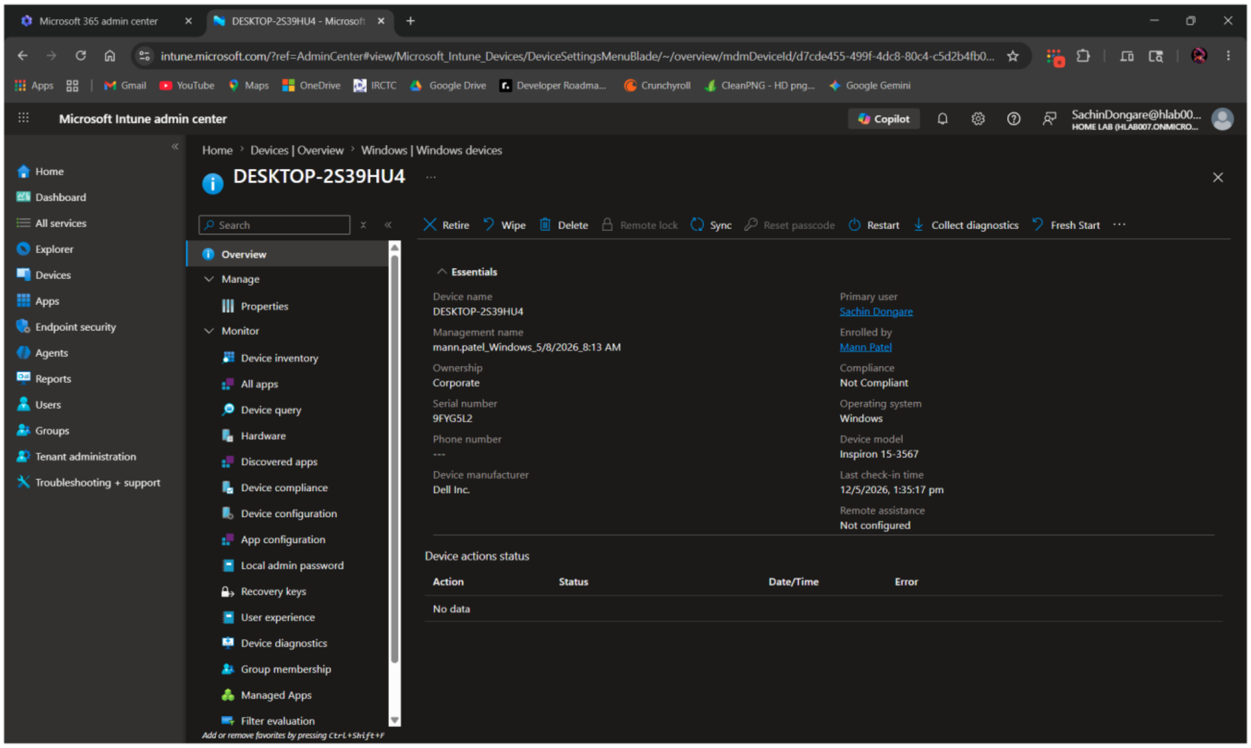


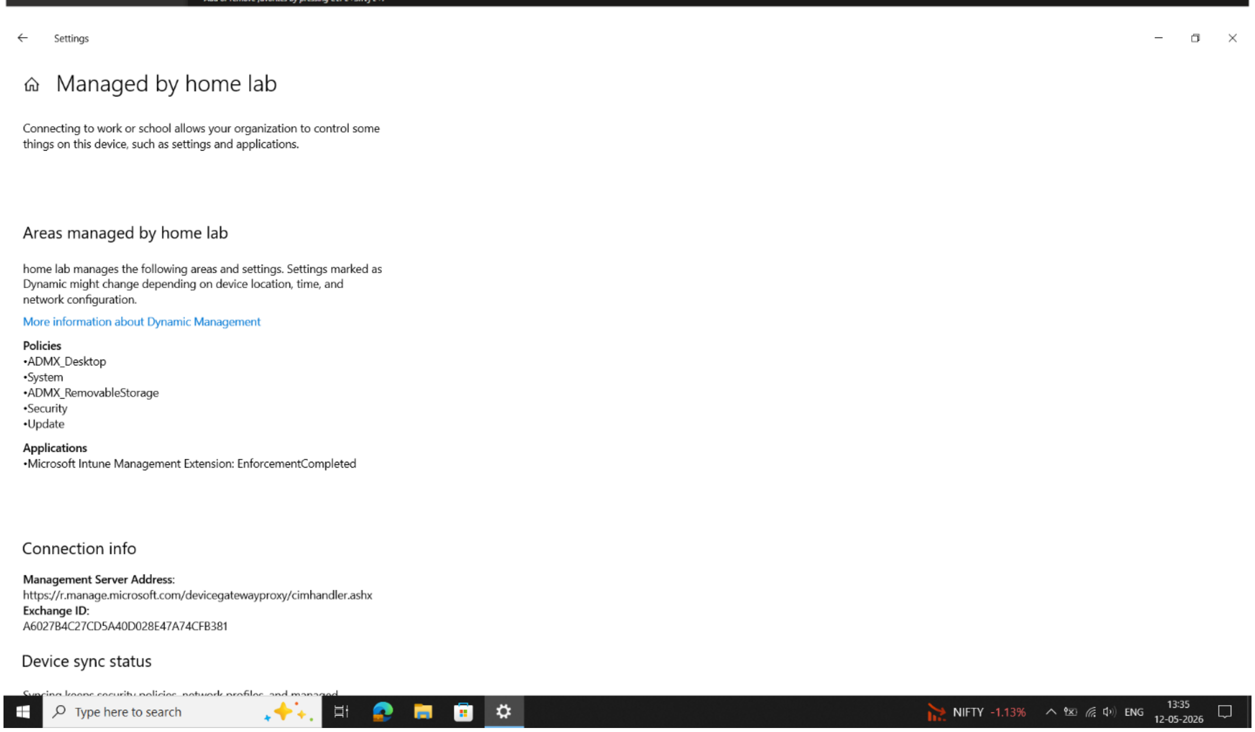
2.2 Microsoft Intune — Device Management

Devices were enrolled into Intune via Entra ID join. Compliance policies, configuration profiles, and app deployments were configured to manage and secure enrolled devices.

Device Enrollment

The test device was enrolled into Intune via Entra ID join. Once enrolled, the device appeared in the Intune portal and was visible for policy assignment and monitoring.

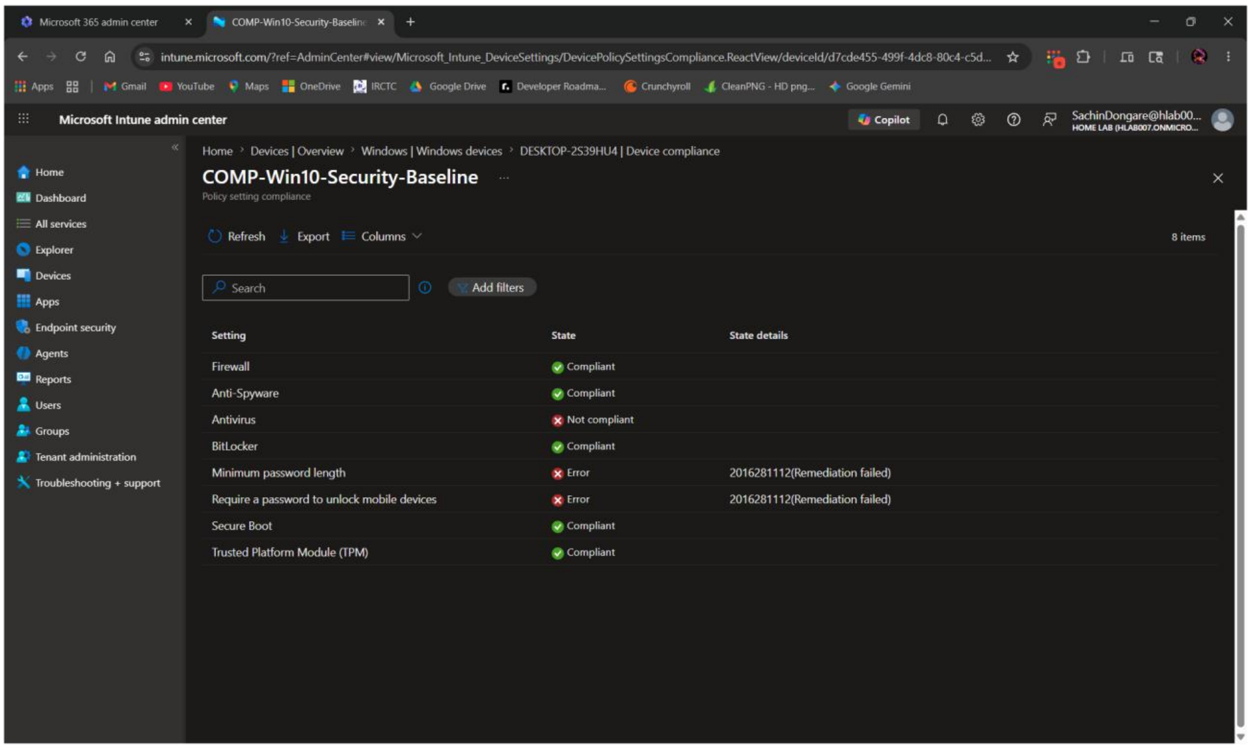




Compliance Policy

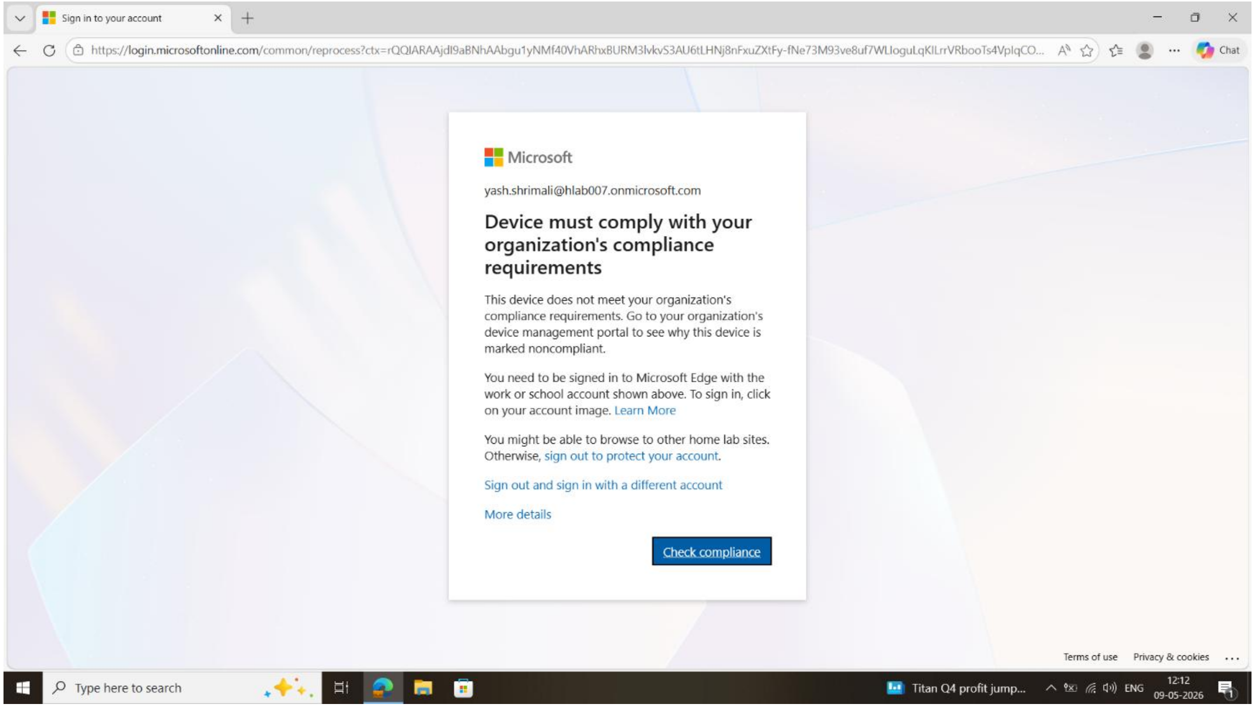
A device compliance policy was configured with the following requirements:

- BitLocker encryption must be enabled
- Minimum Windows version enforced
- Antivirus must be active and reporting
- Minimum password length requirement



Non-Compliant Device — Email Block

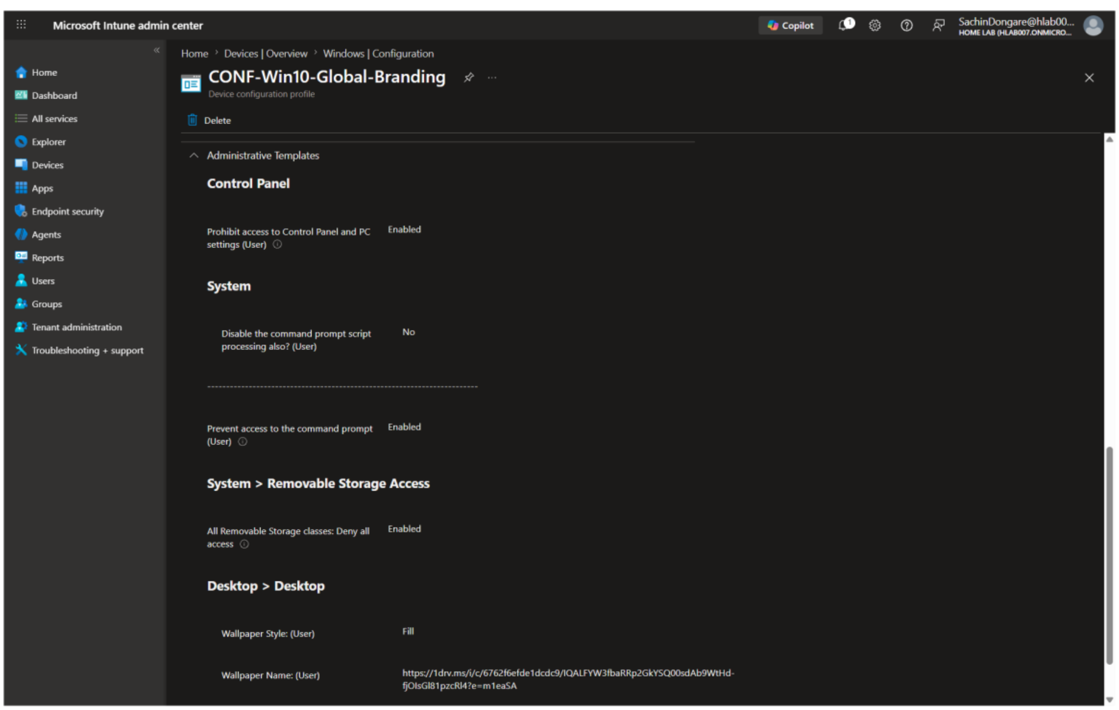
A Conditional Access policy was configured to block non-compliant devices from accessing company email. When the test device was marked non-compliant, access to the mailbox was blocked — demonstrating end-to-end enforcement of device compliance.



Configuration Profiles

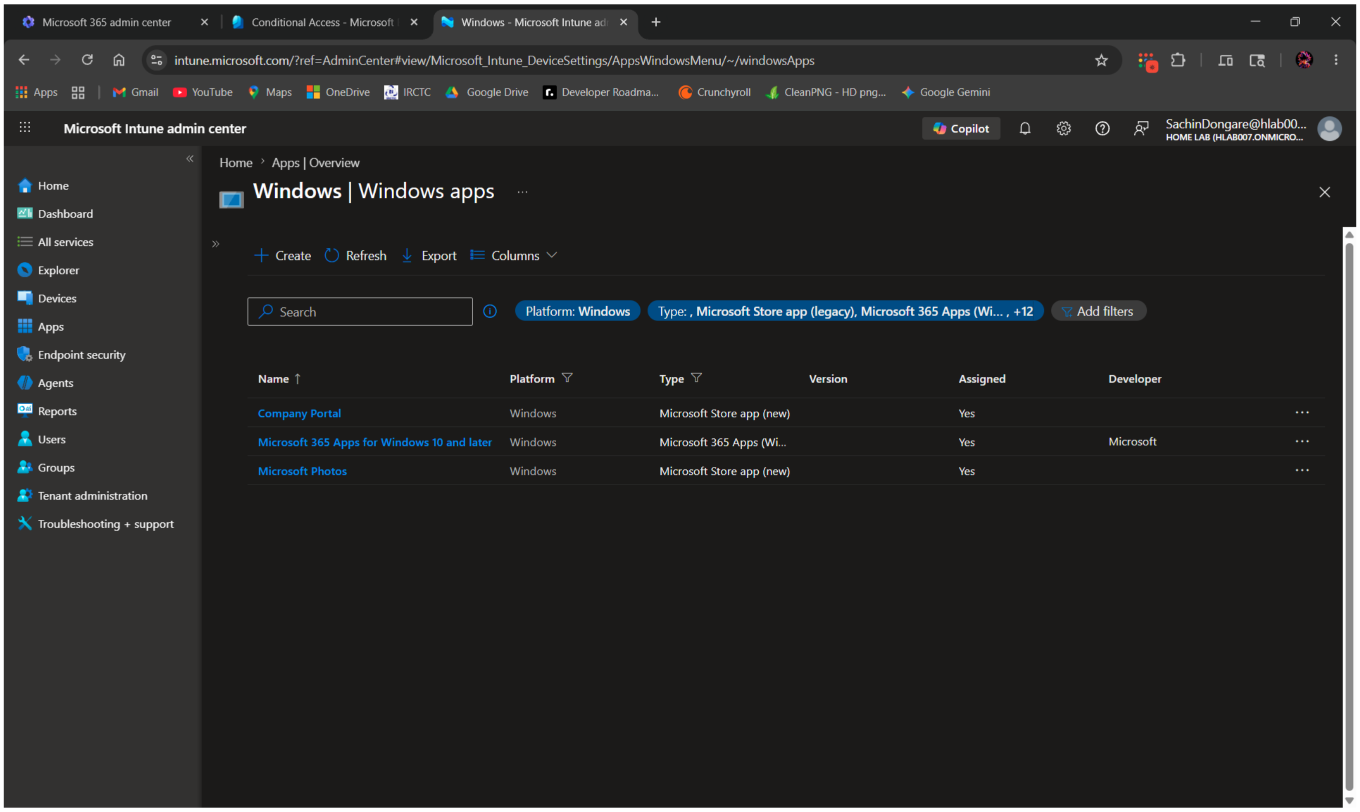
Configuration profiles were created and deployed to enforce the following settings on managed devices:

- USB storage blocked to prevent unauthorized data transfer
- Command Prompt (CMD) access restricted
- Control Panel access blocked for standard users



App Deployment

Microsoft 365 Apps and Company Portal were deployed to enrolled devices via Intune app deployment. Apps were assigned to user groups and installed automatically without any manual intervention on the device.

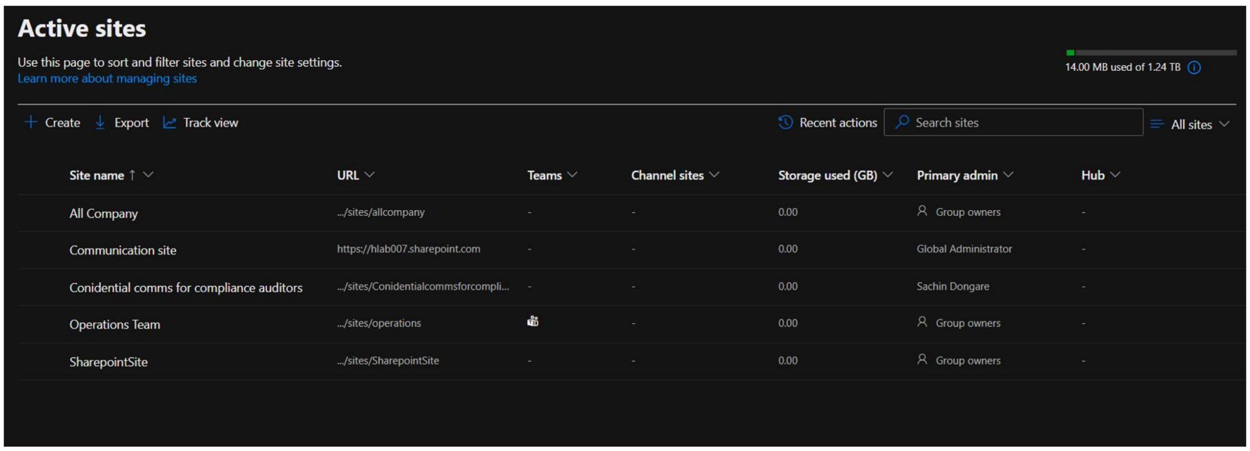


2.3 SharePoint Online

SharePoint sites were created and configured to demonstrate site management, permissions, and access control in a Microsoft 365 environment.

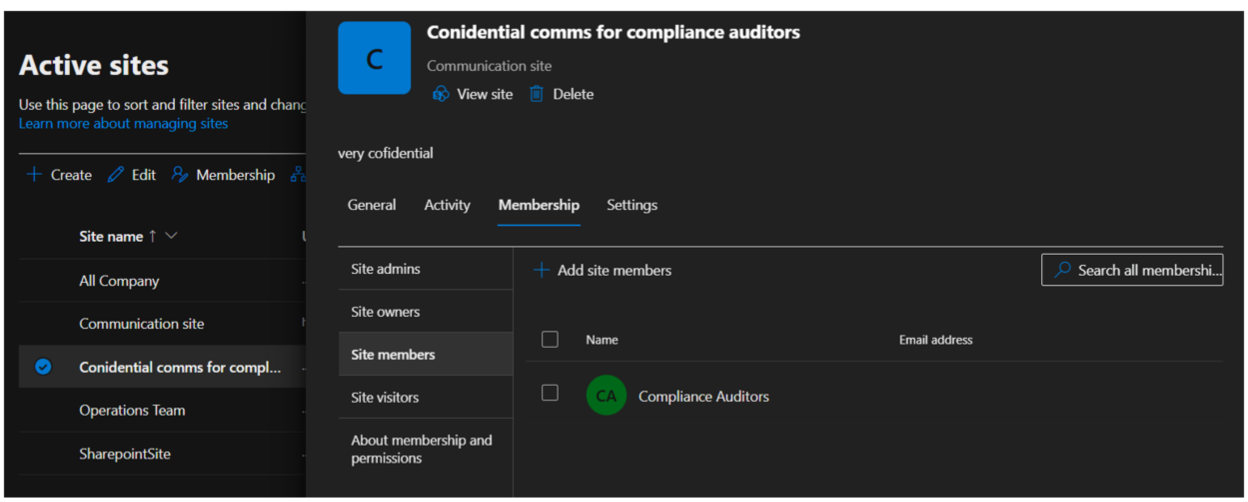
Team Site and Communication Site

Both a Team Site and a Communication Site were created. The Team Site was linked to a Microsoft 365 Group, automatically provisioning a shared workspace. The Communication Site was created as a standalone portal for broadcasting information to a wider audience.



Site Permissions

Permissions were configured on the Communication Site by assigning the previously created Security Group as Members. This demonstrates how group-based access control works in SharePoint — access is managed through group membership rather than individual users.

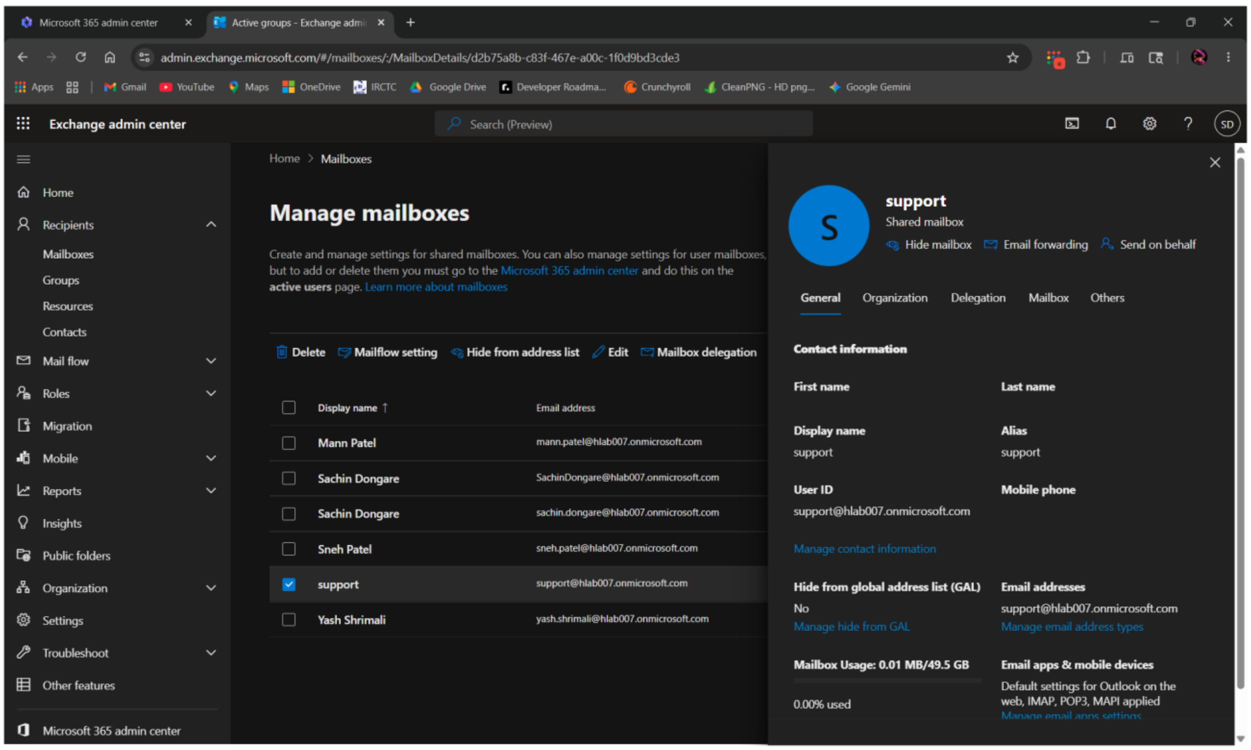


2.4 Exchange Online

Exchange Online was configured to demonstrate mailbox management and mail flow rule implementation.

Shared Mailbox

A shared mailbox was created and team members were added with Full Access and Send As permissions. This allows multiple users to monitor and reply from a single email address — commonly used for support or team inboxes.

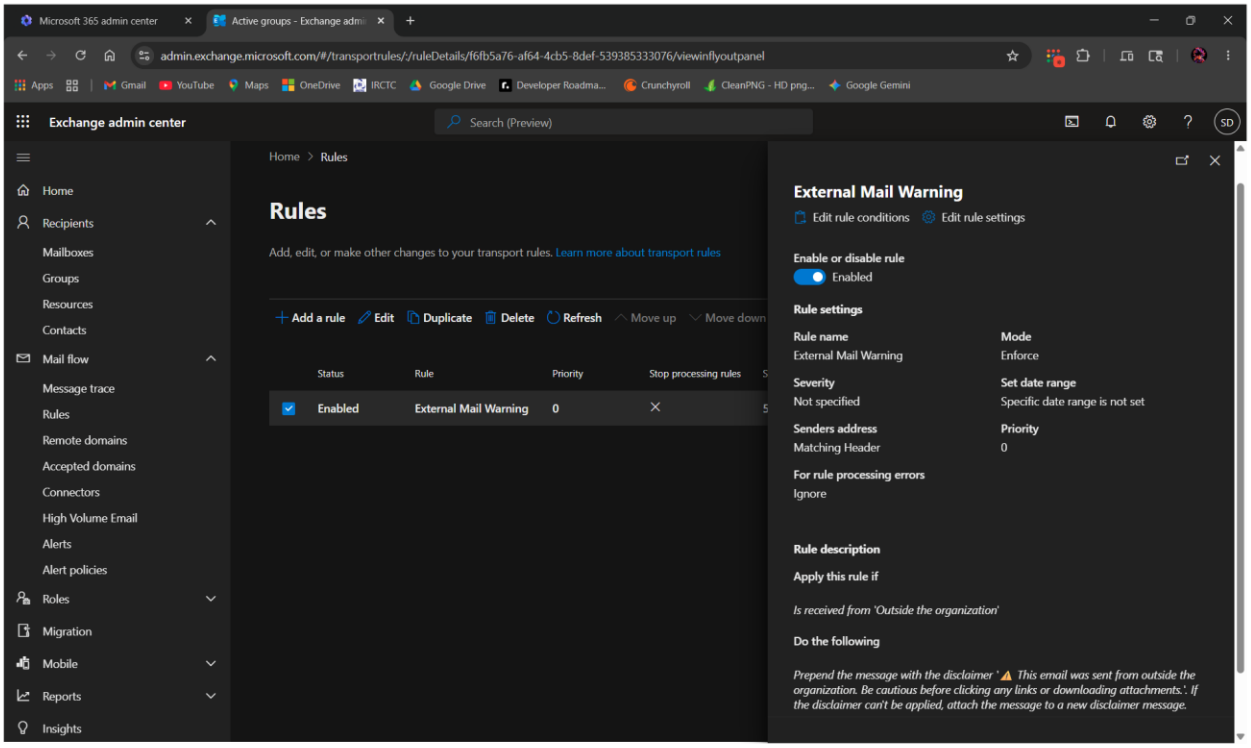


Mail Flow Rule — External Email Warning

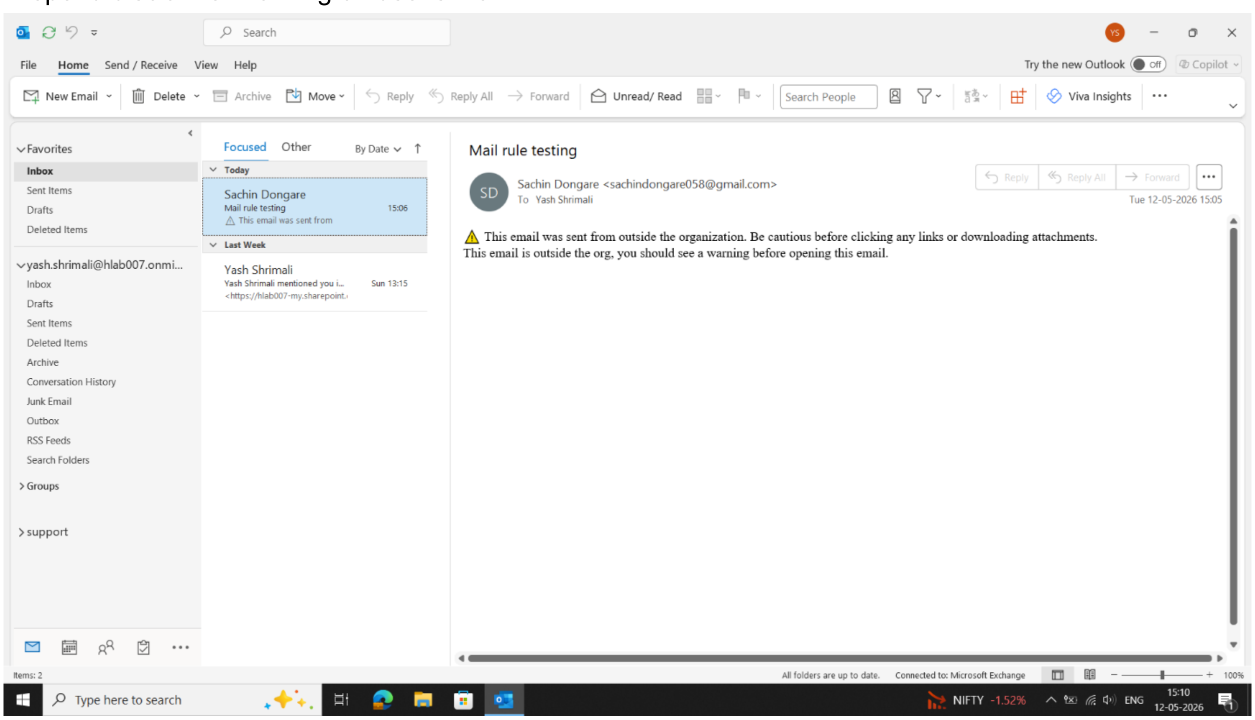
A mail flow rule was configured to automatically prepend a warning banner to any email received from outside the organization. This alerts employees to be cautious with external emails and helps reduce the risk of phishing attacks.

- Condition: Sender is located outside the organization
- Action: Prepend disclaimer warning banner to the email

The rule was tested by sending an email from a personal Gmail account to the Microsoft 365 mailbox. The warning banner appeared at the top of the received email confirming the rule worked correctly.



Prepend disclaimer warning on user's mail.



Conclusion

This home lab covered the core areas of enterprise IT administration — from on-premise Active Directory and Group Policy to cloud-based Microsoft 365 services including Entra ID, Intune, Exchange Online, and SharePoint Online.

Every configuration in this report was implemented hands-on, tested, and verified on real or virtualized hardware. The lab was built to develop practical skills that translate directly to IT support and system administration roles.